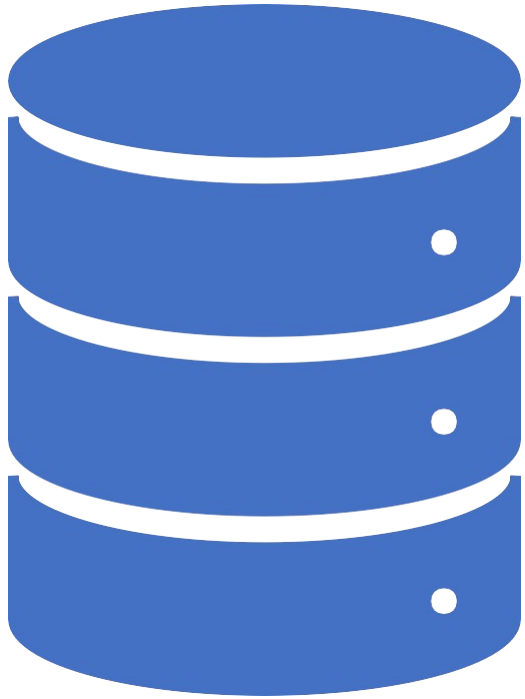
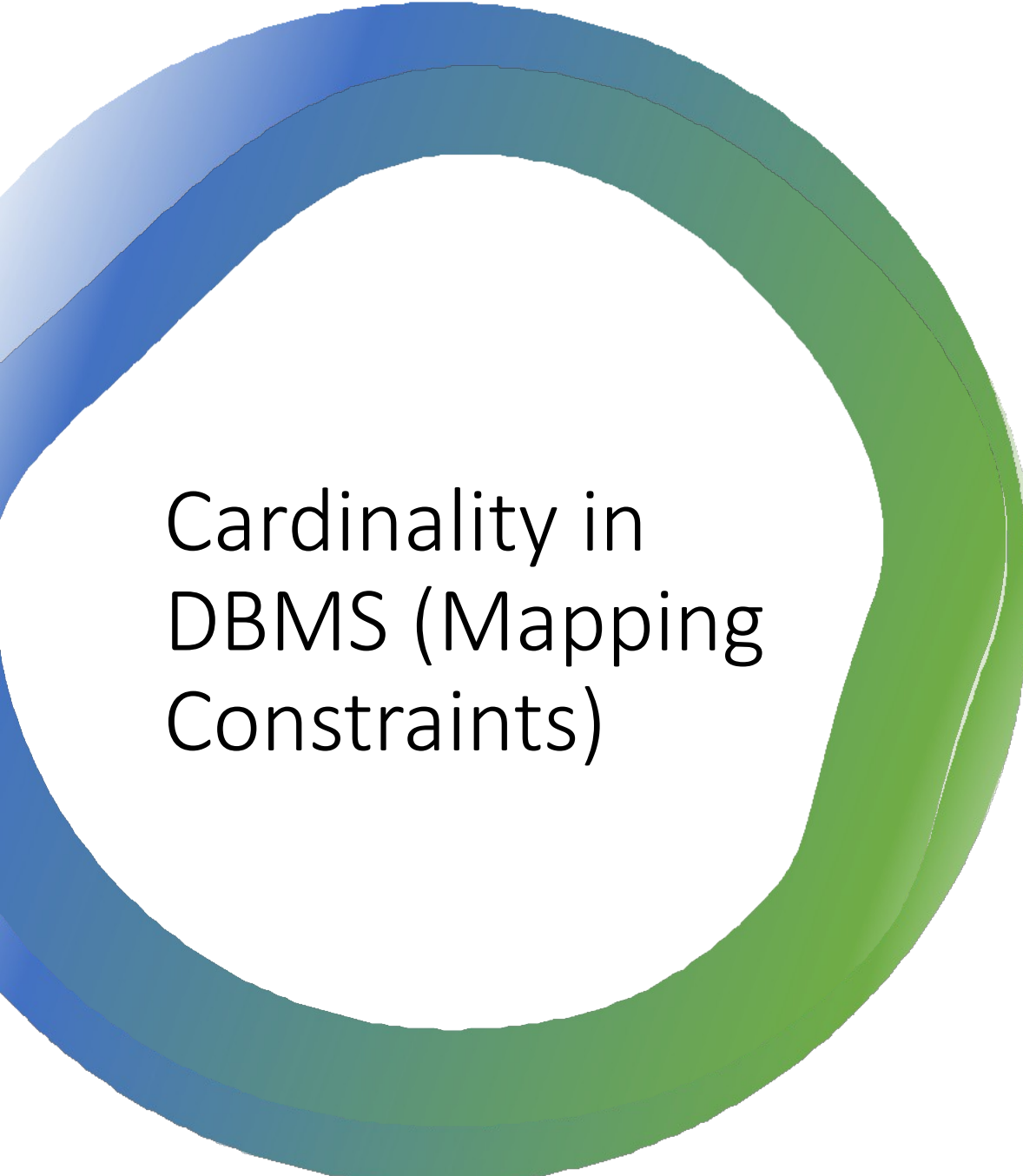


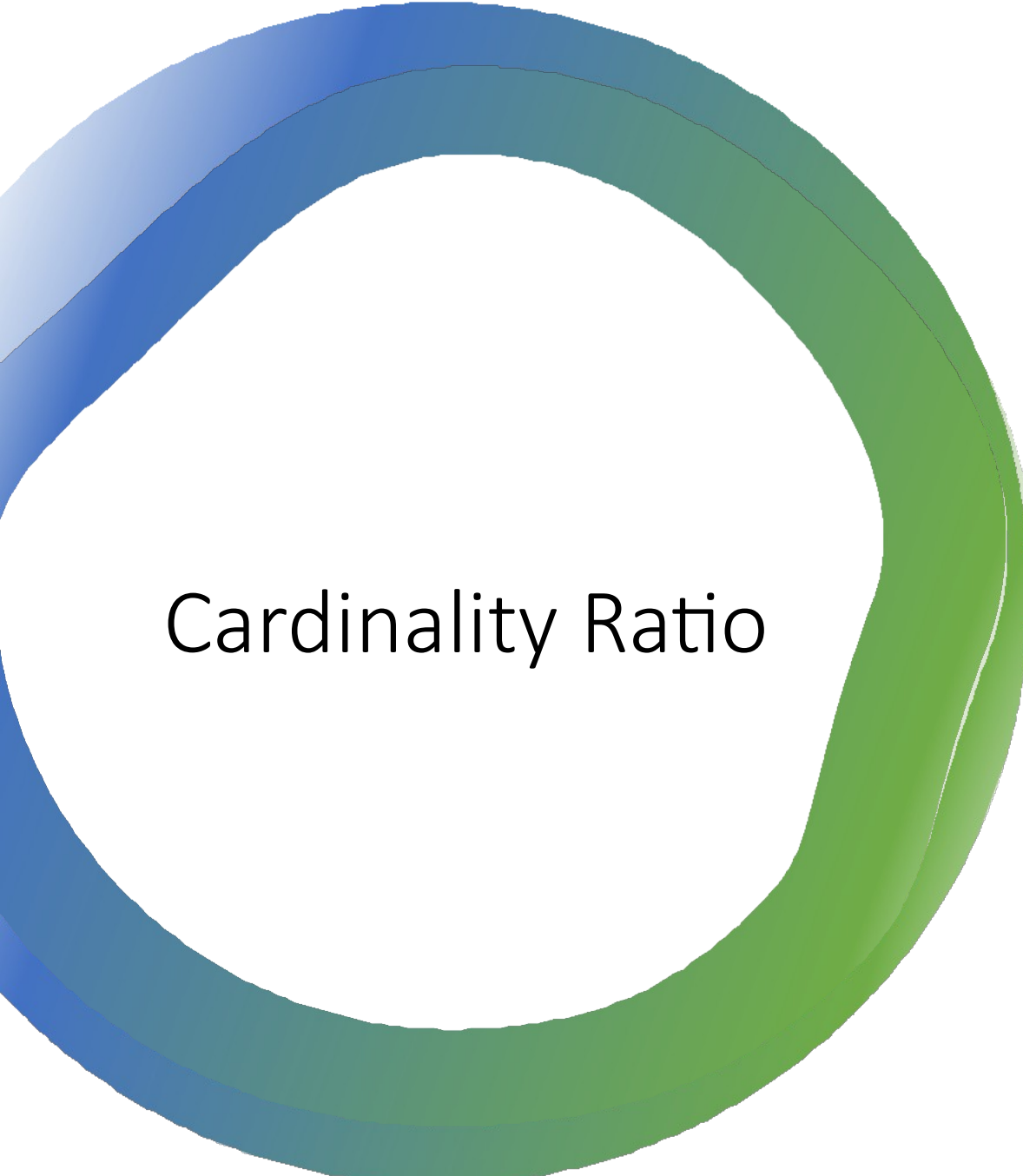


Database System
[*Mapping Constraints in
DBMS*]



Cardinality in DBMS (Mapping Constraints)

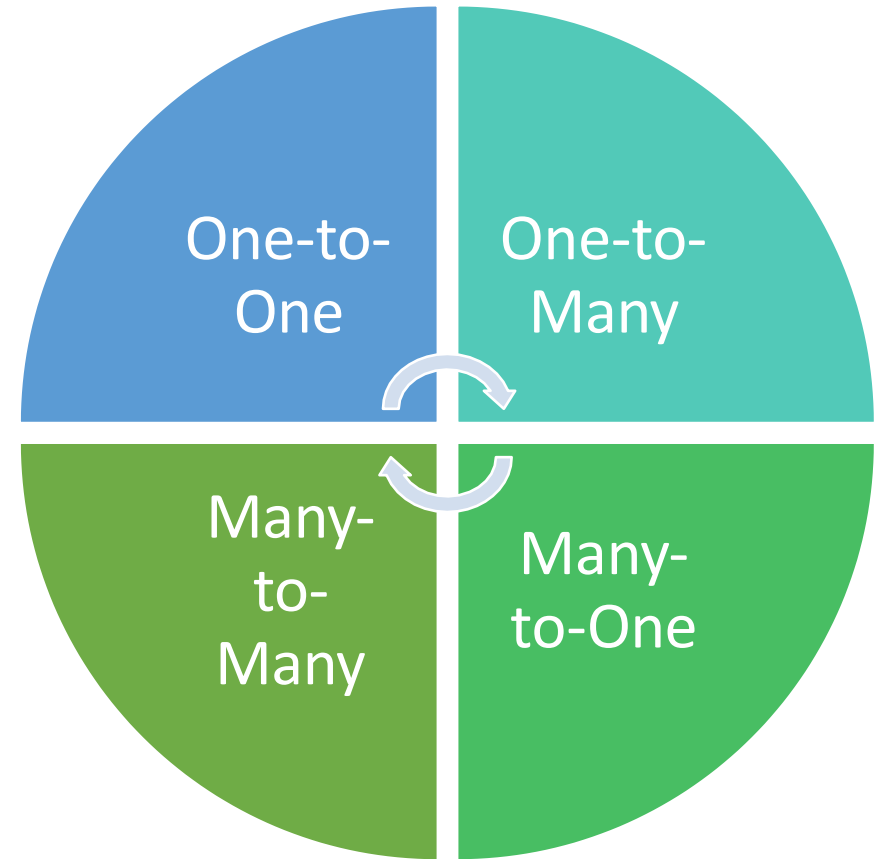
- **Cardinality** means *how the entities are arranged to each other* or *what is the relationship structure between entities in a relationship set*.
- In a DBMS, Cardinality represents a ***number that denotes how many times an entity is participating with another entity*** in a relationship set.
- The Cardinality of DBMS is a very important attribute in representing the structure of a Database.
- In a table, the number of rows or tuples represents the Cardinality.

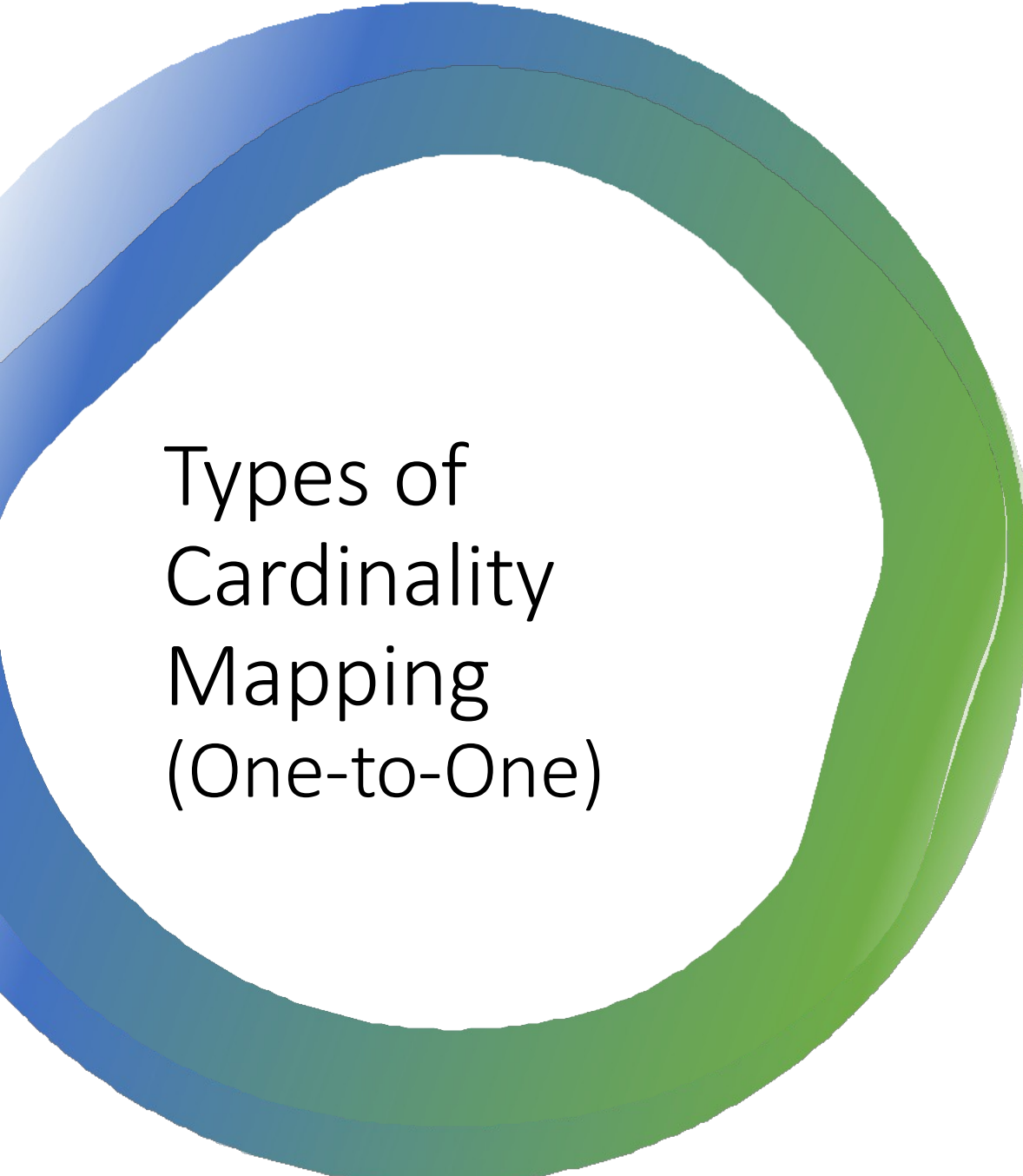


Cardinality Ratio

- Cardinality ratio is also called **Cardinality Mapping**, which represents the mapping of one entity set to another entity set in a relationship set.
- We generally take the example of a binary relationship set where two entities are mapped to each other.
- Cardinality is very important in the Database of various businesses.
 - For example, if we want to track the purchase history of each customer then we can use the one-to-many cardinality to find the data of a specific customer.
- The Cardinality model can be used in Databases by Database Managers for a variety of purposes, but corporations often use it to evaluate customer or inventory data.

Types of Cardinality Mapping





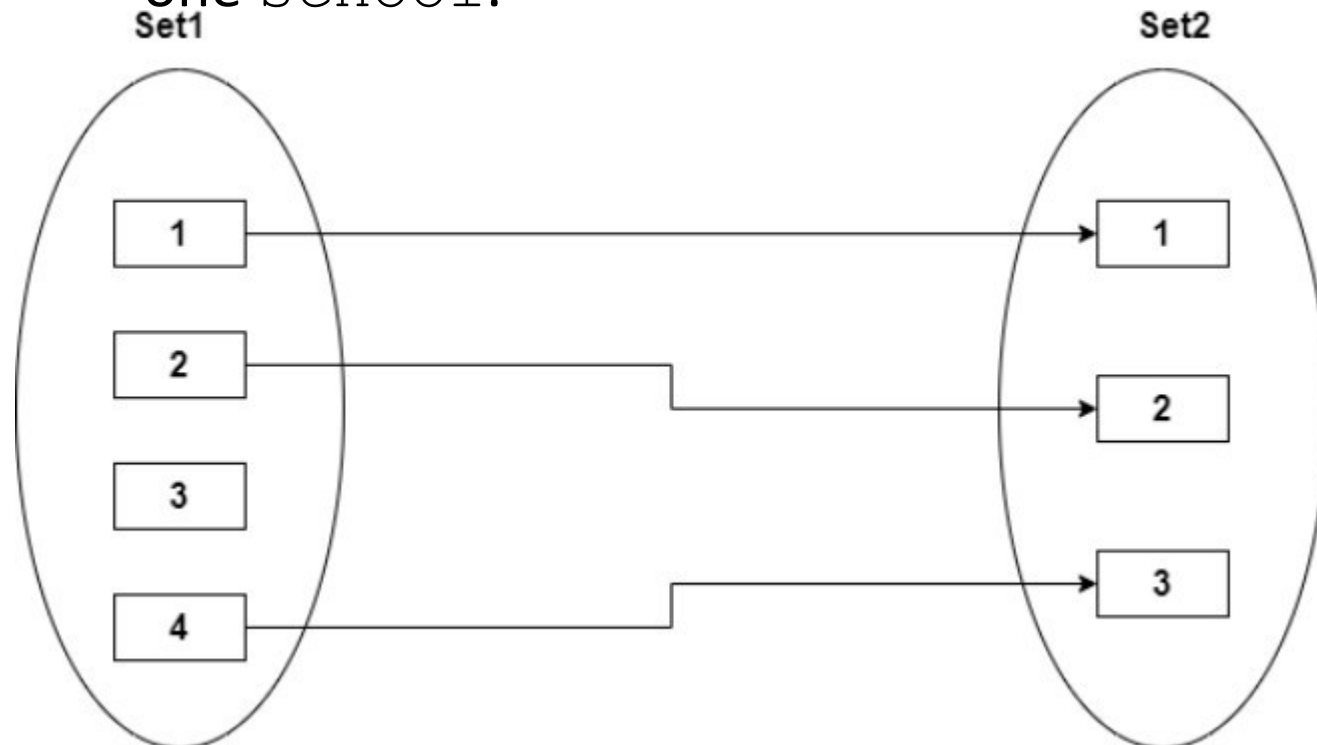
Types of Cardinality Mapping (One-to-One)

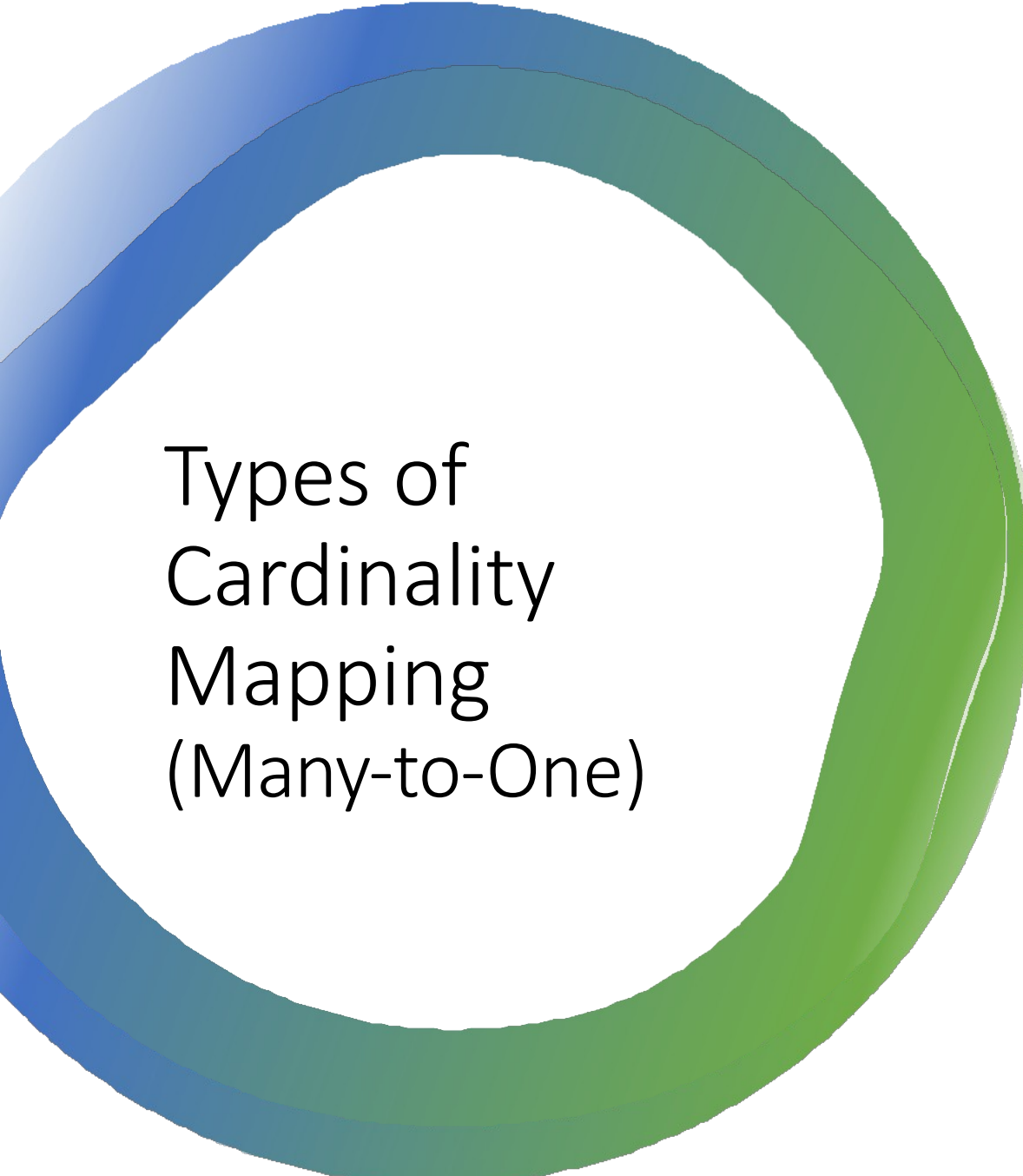
- One to one cardinality is represented by a **1:1** symbol.
- In this, there is *at most one relationship from one entity to another entity*.
- There are a lot of examples of one-to-one cardinality in real life databases.
 - For example, one `student` can have only one `student id`, and one `student id` can belong to only one `student`.
 - So, the relationship mapping between `student` and `student id` will be one to one cardinality mapping.

Types of Cardinality Mapping (One-to-One)

Note: it is not necessary that there would be a mapping for all entities in an entity set in one-to-one cardinality. Some entities cannot participate in the mapping.

- Another example is the relationship between the `director` of the `school` and the `school` because one `school` can have a maximum of one `director`, and one `director` can belong to only one `school`.



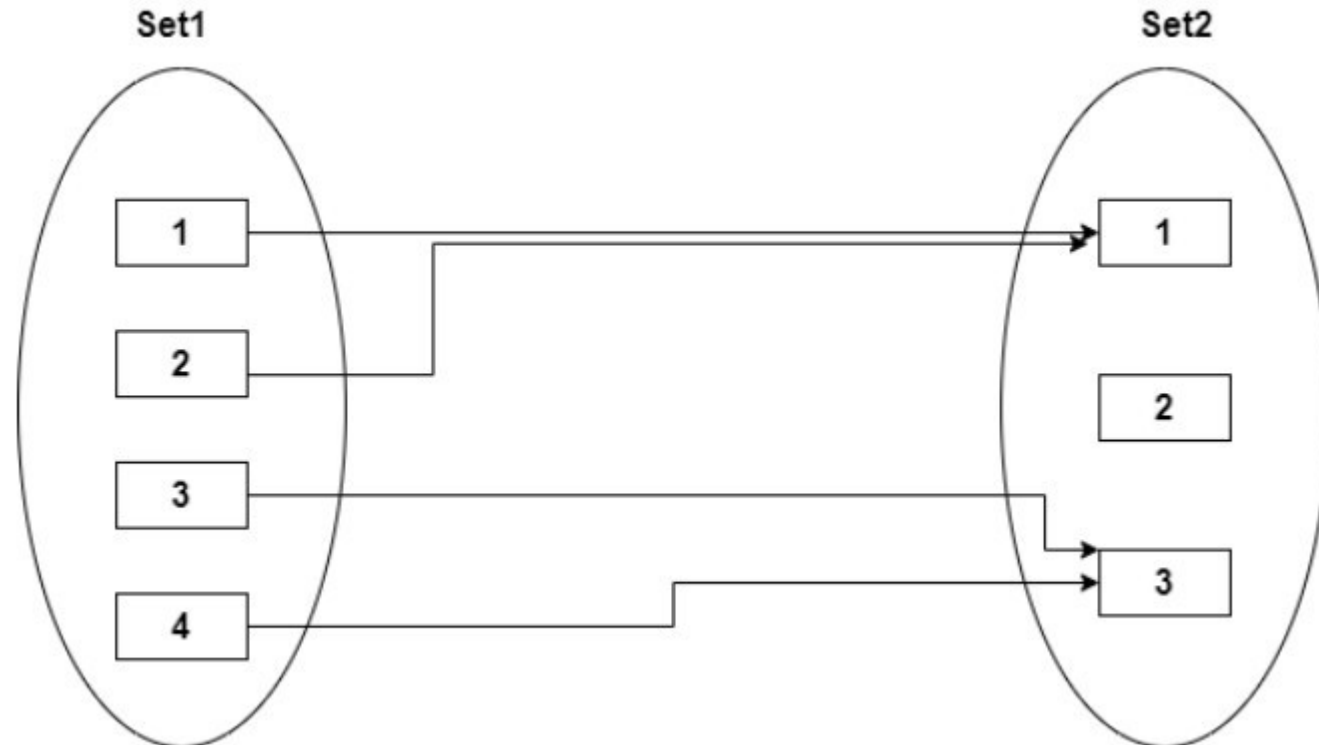


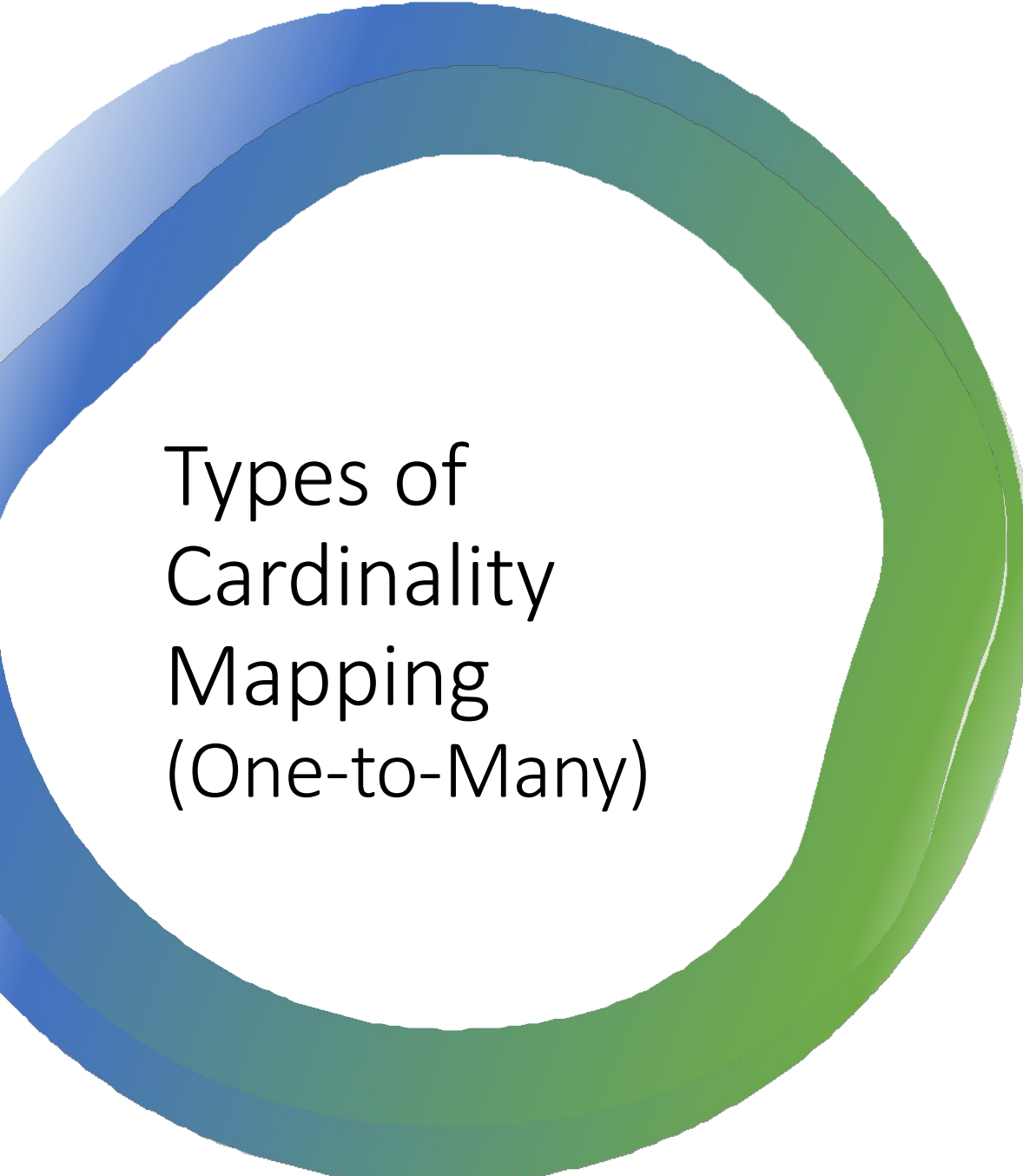
Types of Cardinality Mapping (Many-to-One)

- In many to one cardinality mapping, *from set 1, there can be multiple sets that can make relationships with a single entity of set 2.*
- Or we can also describe it as *from set 2, and one entity can make a relationship with more than one entity of set 1.*
- One to one Cardinality is the subset of Many to one Cardinality.
- It can be represented by **M:1**.

Types of Cardinality Mapping (Many-to-One)

- For example, there are multiple patients in a hospital who are served by a single doctor, so the relationship between patients and doctors can be represented by Many to one Cardinality.



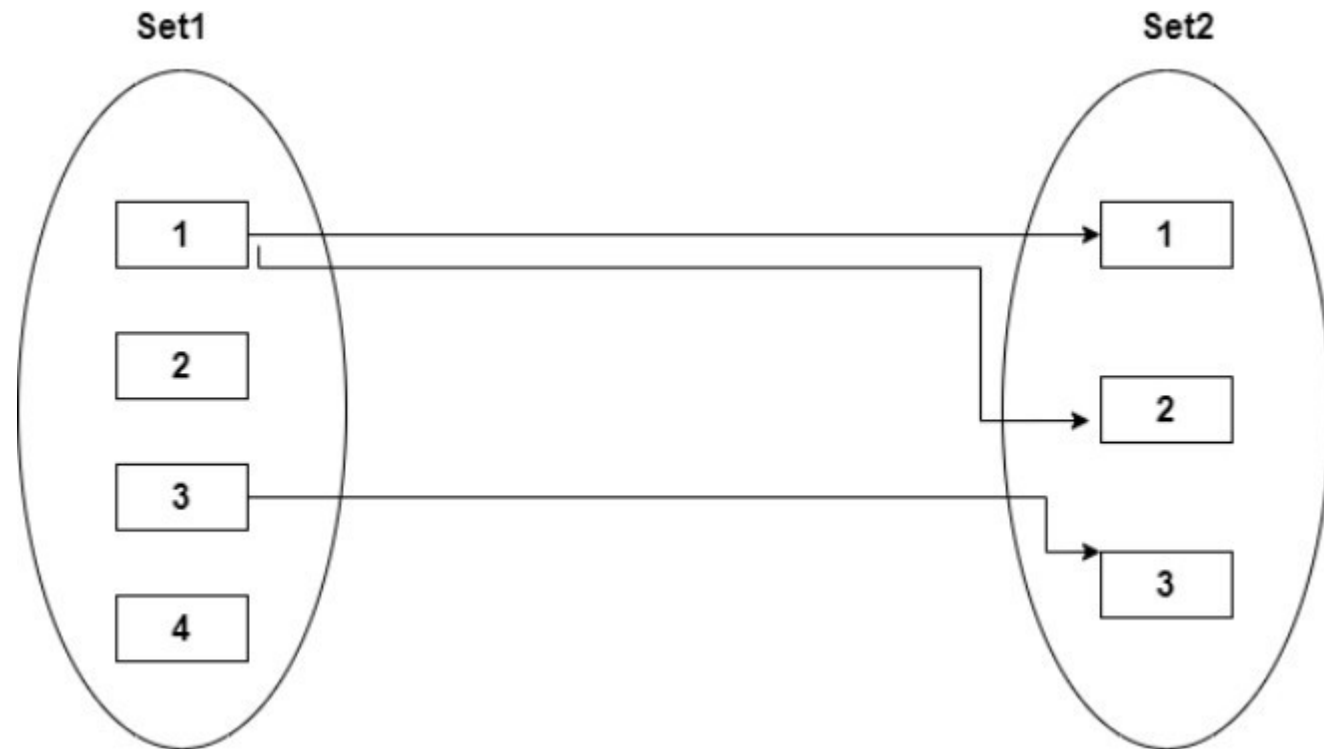


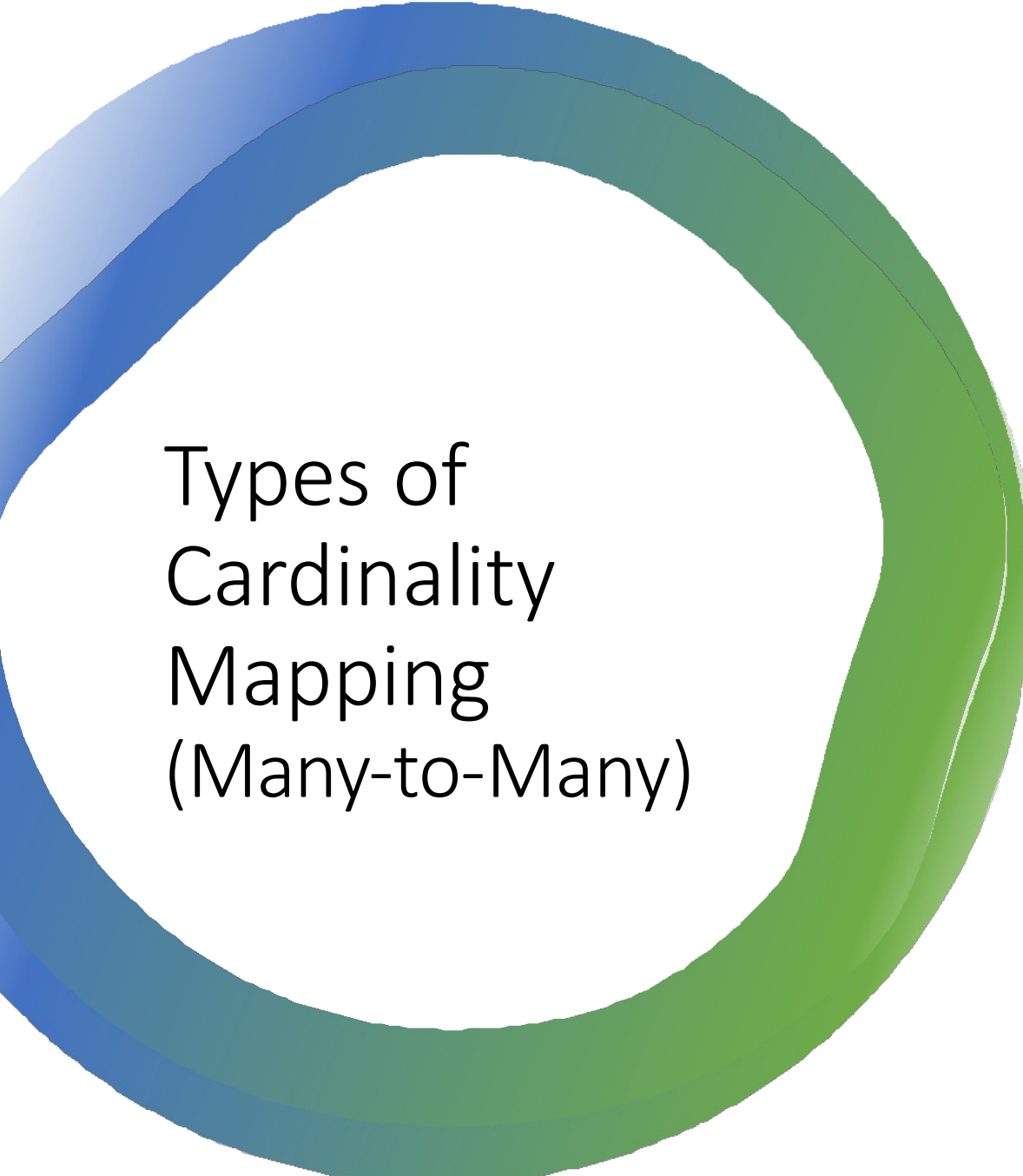
Types of Cardinality Mapping (One-to-Many)

- In One-to-many cardinality mapping, *from set 1, there can be a maximum single set that can make relationships with a single or more than one entity of set 2.*
- Or we can also describe it as *from set 2, more than one entity can make a relationship with only one entity of set 1.*
- One to one cardinality is the subset of One-to-many Cardinality.
- It can be represented by **1: M**.

Types of Cardinality Mapping (One-to-Many)

- For Example, in a hospital, there can be various compounders, so the relationship between the hospital and compounders can be mapped through One-to-many Cardinality.



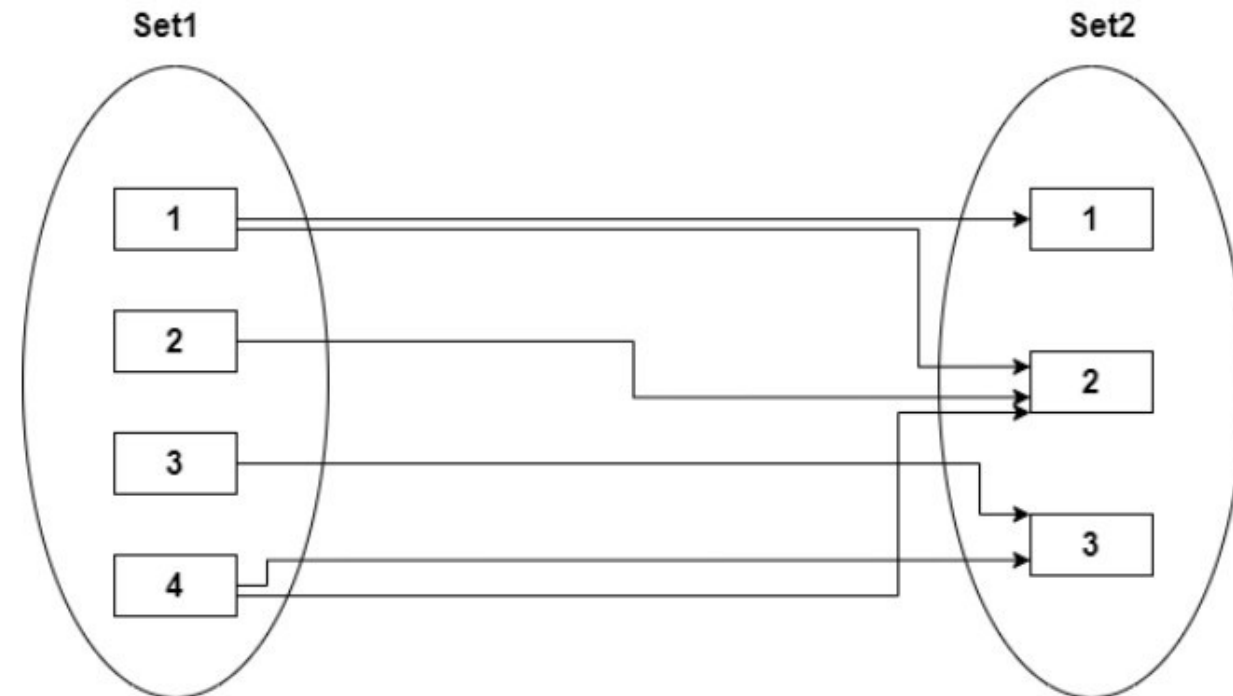


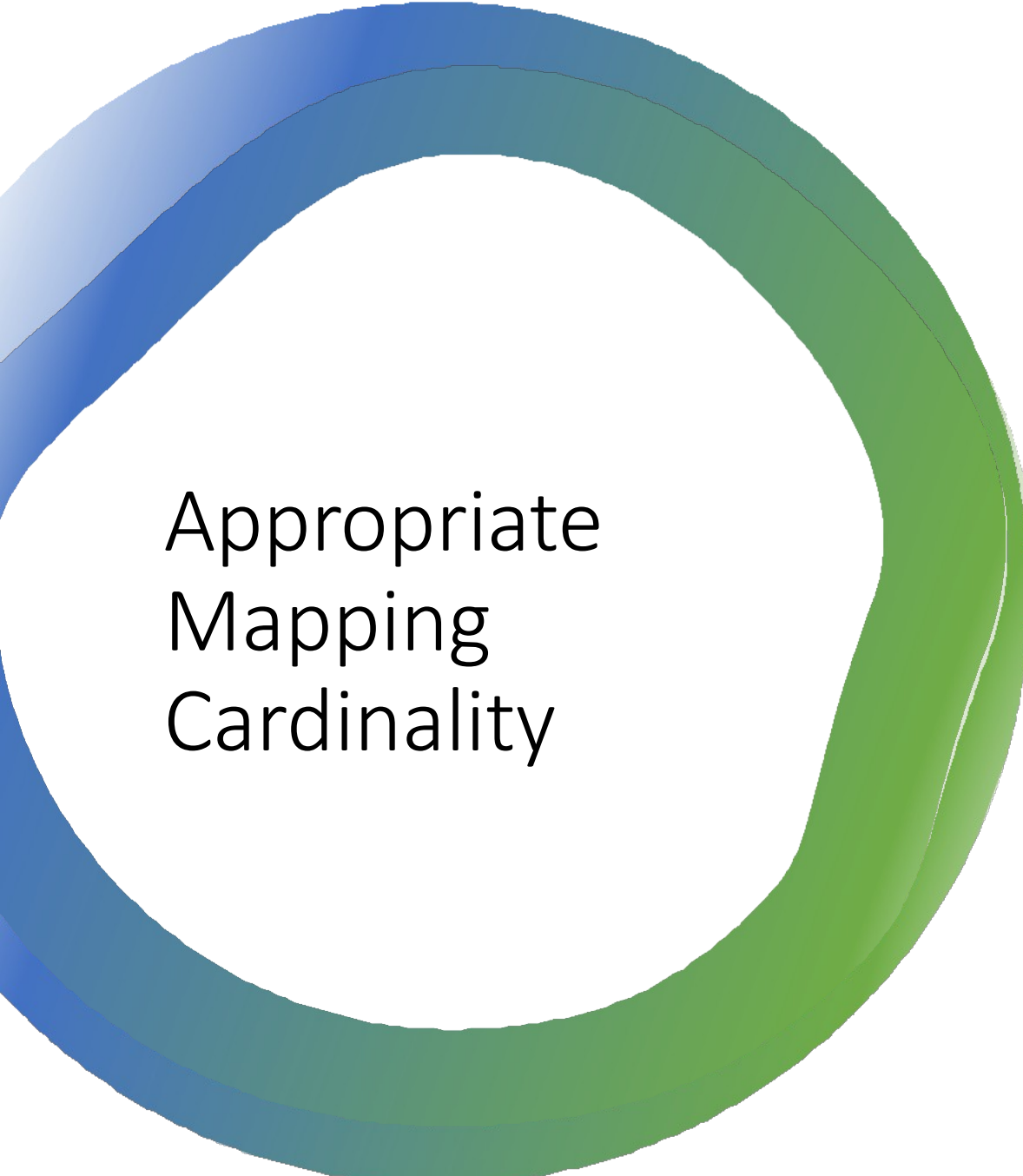
Types of Cardinality Mapping (Many-to-Many)

- In many-to-many cardinalities mapping, there can be *one or more than one entity that can associate with one or more than one entity of set 2.*
- In the same way *from the end of set 2, one or more than one entity can make a relation with one or more than one entity of set 1.*
- It is represented by **M: N** or **N: M**.
- One to one cardinality, One to many cardinalities, and Many to one cardinality is the subset of the many to many cardinalities.

Types of Cardinality Mapping (Many-to-Many)

- For Example, in a college, multiple students can work on a single project, and a single student can also work on multiple projects.
- So, the relationship between the project and the student can be represented by many to





Appropriate Mapping Cardinality

- Evidently, the real-world context in which the relation set is modeled determines the Appropriate Mapping Cardinality for a specific relation set.
 - We can combine relational tables with many involved tables if the Cardinality is one-to-many or many-to-one.
 - One entity can be combined with a relation table if it has a one-to-one relationship and total participation, and two entities can be combined with their relation to form a single table if both of them have total participation.
 - We cannot mix any two tables if the Cardinality is many-to-many.